# AI CRASH DETECTION SOFTWARE

Presented By,
TEAM 4

# TEAM MEMBERS

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# Project Overview

- Cutting-edge crash detection AI system
- Enhanced accuracy and responsiveness
- Multi-platform compatibility (PC & Mobile)
- Real-time crash detection & emergency response

# SYSTEM DESIGN DETAILS

**Development Tools & Technologies**

- **IDE:**
  - PyCharm for PC application
  - Android Studio for mobile app development
- **Languages:**
  - Python for PC-based application
  - Dart (Flutter framework) for mobile application
  - Java for Android-specific implementations
- **Backend Infrastrcture:**
  - Firebase for database management, real-time synchronization, and app hosting
  - Firebase Cloud Messaging (FCM) for notifications
- **Frontend:**
  - Tkinter or CustomTkinter for PC GUI
  - Flutter for mobile app interface

# AI FRAMEWORKS & LIBRARIES

**AI Model Development**

- **Ultralytics YOLOv8**:
  - State-of-the-art object detection model
  - Optimized for crash detection scenarios
- **Torch & Torchvision:**
  - Core PyTorch framework used for training and deployment
- **OpenCV-python-headless:**
  - Efficient library for real-time image and video stream processing
- **Dataset:**
  - Over 8,600 labelled images and videos for robust crash event detection
  - Diverse scenarios to enhance model generalization

# KEY FEATURES

## System Features

**1.Real-Time Crash Detection:**

- •Processes video feeds to identify crash events immediately
- •High accuracy with minimal false positives and negatives

**2.Automated Notifications:**

- •Sends alerts to emergency responders via Firebase Cloud Messaging
- •Ensures timely action in critical situations

**3.Cross-Platform Integration:**

- •Synchronization between PC and mobile applications
- •Unified data and seamless monitoring experience

**4.Image & Video Processing:**

- •Handles high-resolution streams efficiently using OpenCV

**5.Performance Analytics:**

- •Tracks detection accuracy, system responsiveness, and user engagement
- •Provides insights for continuous optimization



# ADVANTAGES

## System Advantages

- **Efficiency:** Real-time operation minimizes delays in emergency response
- **Scalability:** Supports expanding datasets and system complexity
- **Accuracy:** Advanced AI models reduce false positives and negatives
- **Cost-Effective:** Cloud-based infrastructure reduces hardware costs
- **Data Security:** Implements robust Firebase security protocols for data privacy
- **User-Friendly:** Intuitive design ensures ease of use for both administrators and end-users

# TECHNOLOGIES & TOOLS

## Technologies Used

**IDE:**

- PyCharm for development and debugging of PC application
- Android Studio for mobile app development

**Frameworks:**

- Flutter for creating cross-platform mobile apps
- PyTorch for building and training AI models

**Backend Infrastructure:**

- Firebase for cloud hosting, database management, and notifications

**Image & Video Processing:**

- OpenCV-python-headless for efficient handling of visual data

**AI Model Version:** YOLOv8 for crash detection

# WORKFLOW DIAGRAM

1
**Data Collection**
Images and videos labeled.

2
**Model Training**
YOLOv8 trained using PyTorch.

3
**Video Stream Input**
Live feeds processed using OpenCV.

4
**Real-time Detection**
AI model identifies crash events.

5
**Notification Trigger**
Firebase sends alerts.

6
**Result Display**
Outputs shown on PC and mobile.

# IMPLEMENTATION DETAILS

**Development:**

Setting up IDEs (PyCharm, Android Studio) and Flutter environment
Training AI models using YOLOv8
Backend integration with Firebase

**Testing:**

Extensive testing with diverse crash scenarios
Performance evaluation and debugging

**Deployment:**

Deployment on PC and mobile platforms
Final validation in real-world environments

# CONCLUSION

**Conclusion**

- Revolutionizing crash detection with AI-driven solutions
- Enhanced real-time responsiveness and multi-platform usability
- Scalable system ready for future advancements
- Aimed at improving emergency response efficiency and saving lives

# THANK YOU